How broad is that claim?

Mapping ***Generalisation*** in NLP Research

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4. **LLMs** are reliable for annotation. ***Generic!***

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The result did not change.

But the claims *generalise* differently.

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How do NLP researchers generalise their findings?

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Let's talk about *generalisation*

Following Tessler and Goodman [2019], who draw on Hume [1978], we consider generalisation to be the process of forming expectations about unobserved cases from regularities in past experience.

Three common forms [Tessler and Goodman, 2019]:

- **Habitual**: `gpt-oss` usually produces reliable annotations.
- **Causality**: Few-shot prompting results in reliable annotations.
- **Generic**: LLMs are reliable for annotation.

Taxonomy **NLPGenX**

Annotation **NLPGenA**

Dataset **NLPGens**

Analysis

Contributions

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Prerequisite: each sentence reports a study outcome.

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⇒ **Genericity**: these cues rule out a generic reading

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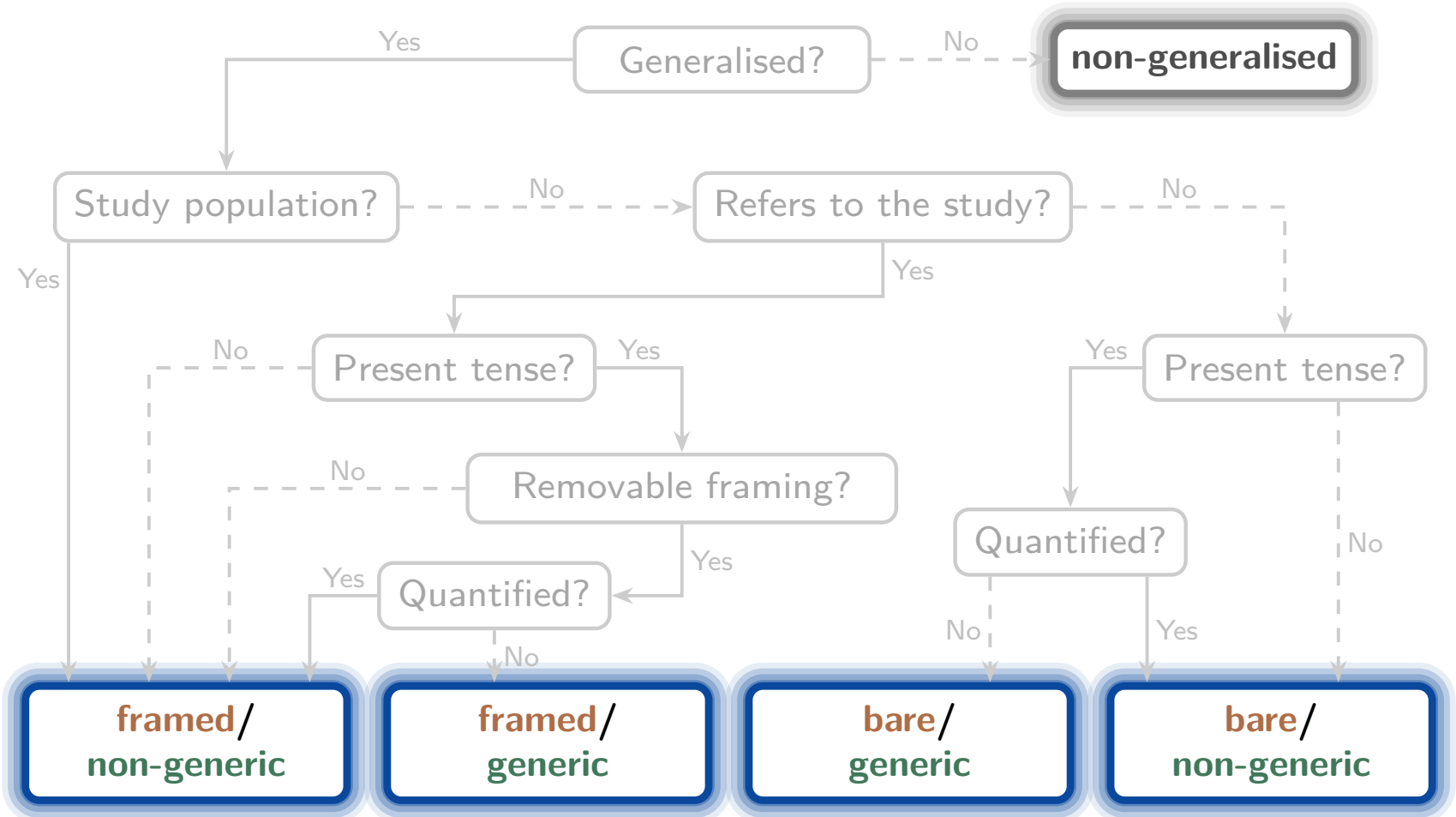
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Genericity: these cues rule out a generic reading

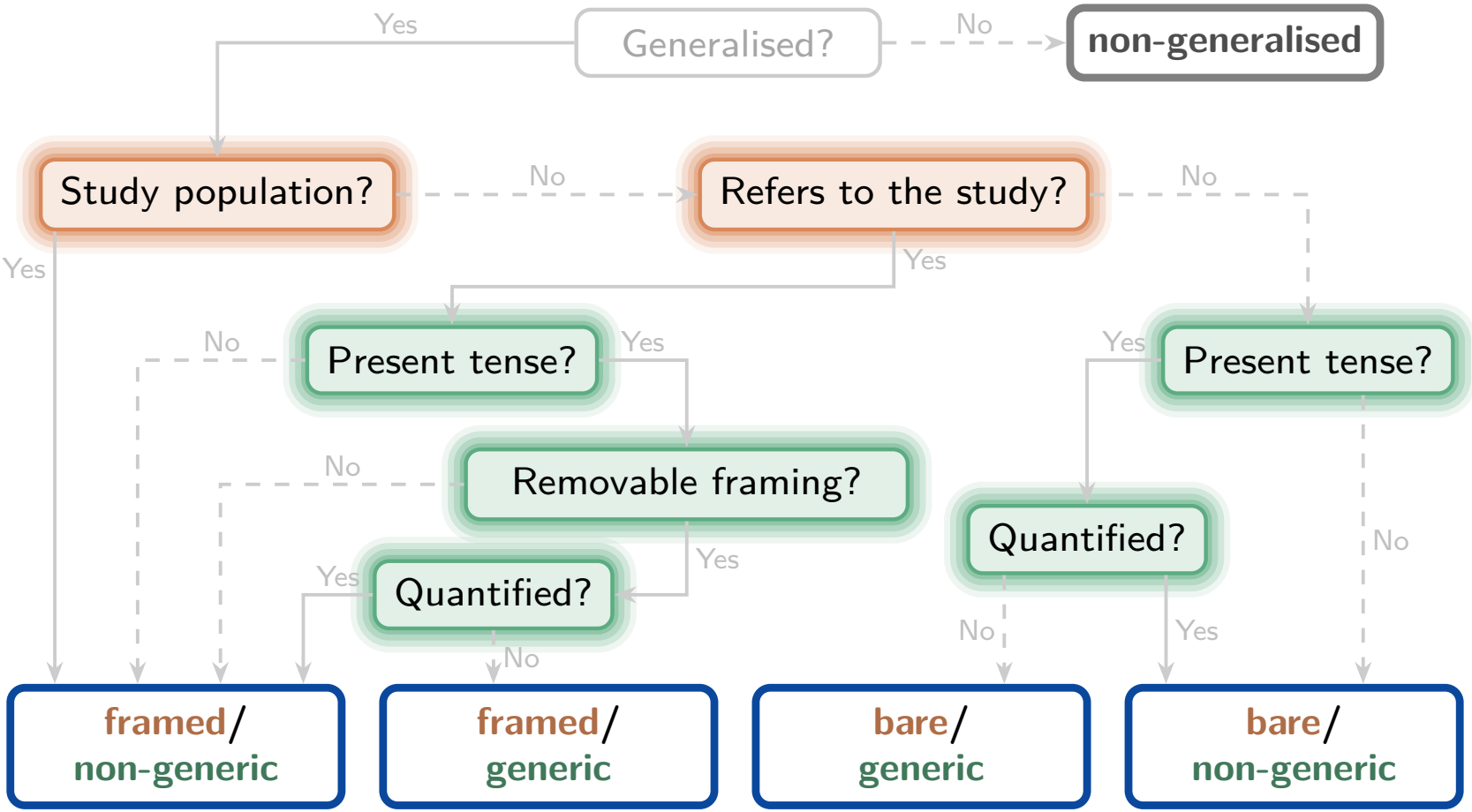
framed/ non-generic	**framed/ generic**
bare/ generic	**bare/ non-generic**

non-generalised

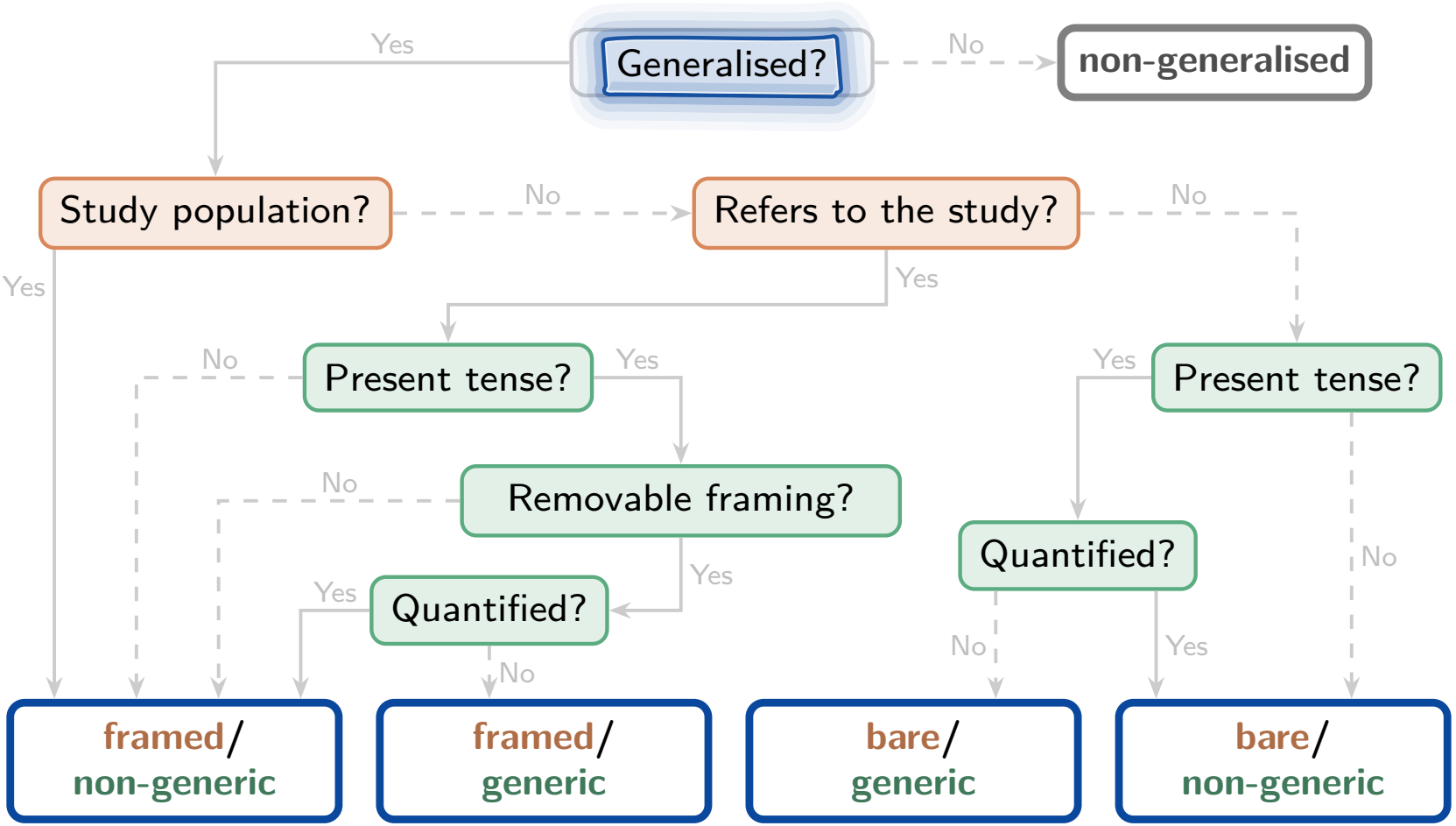
NLPGenA: Annotation framework for NLP generalisation



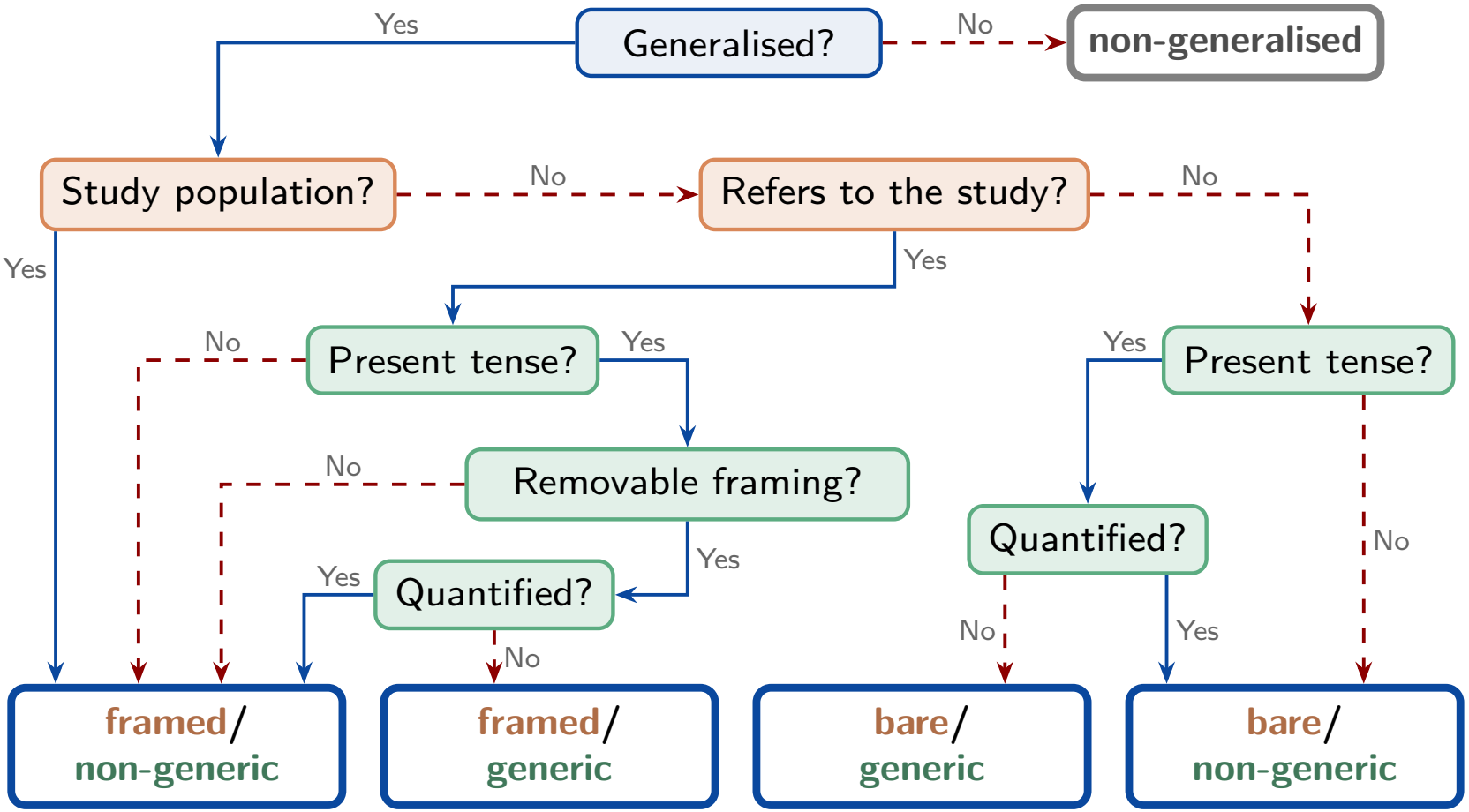
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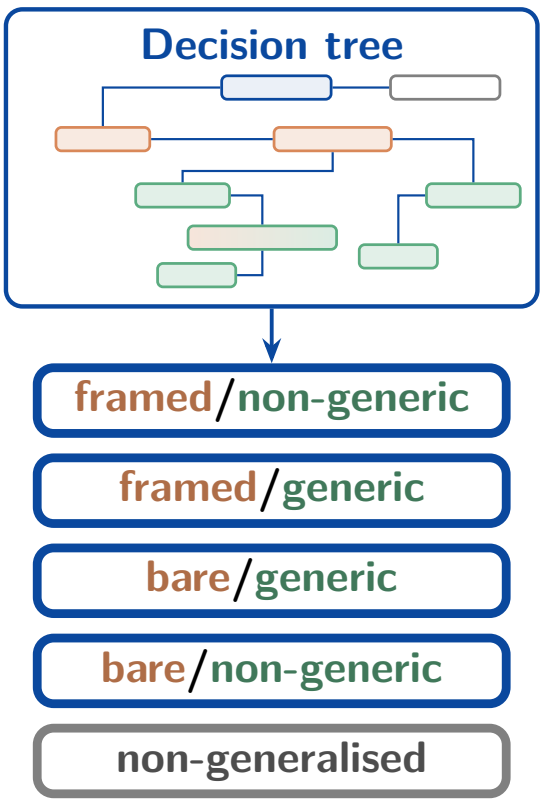
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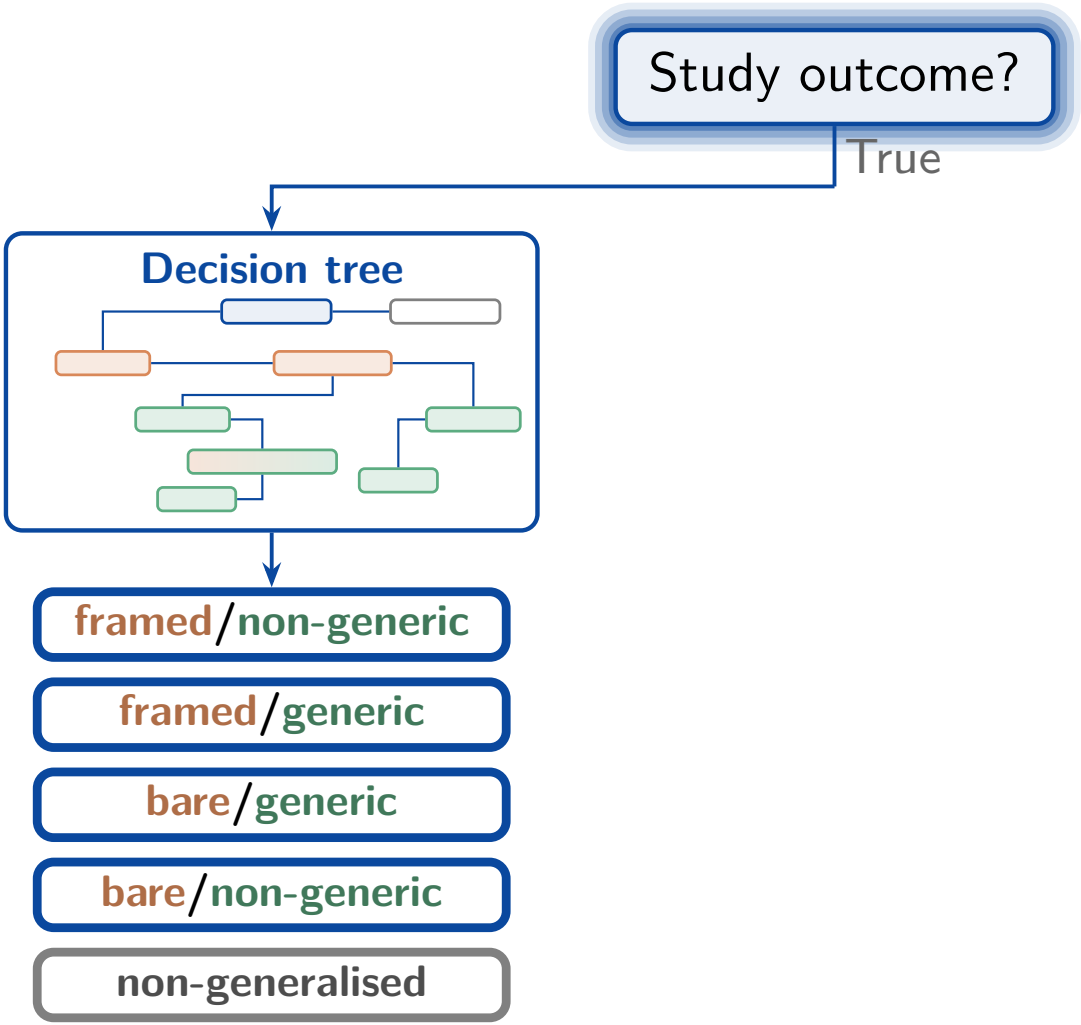
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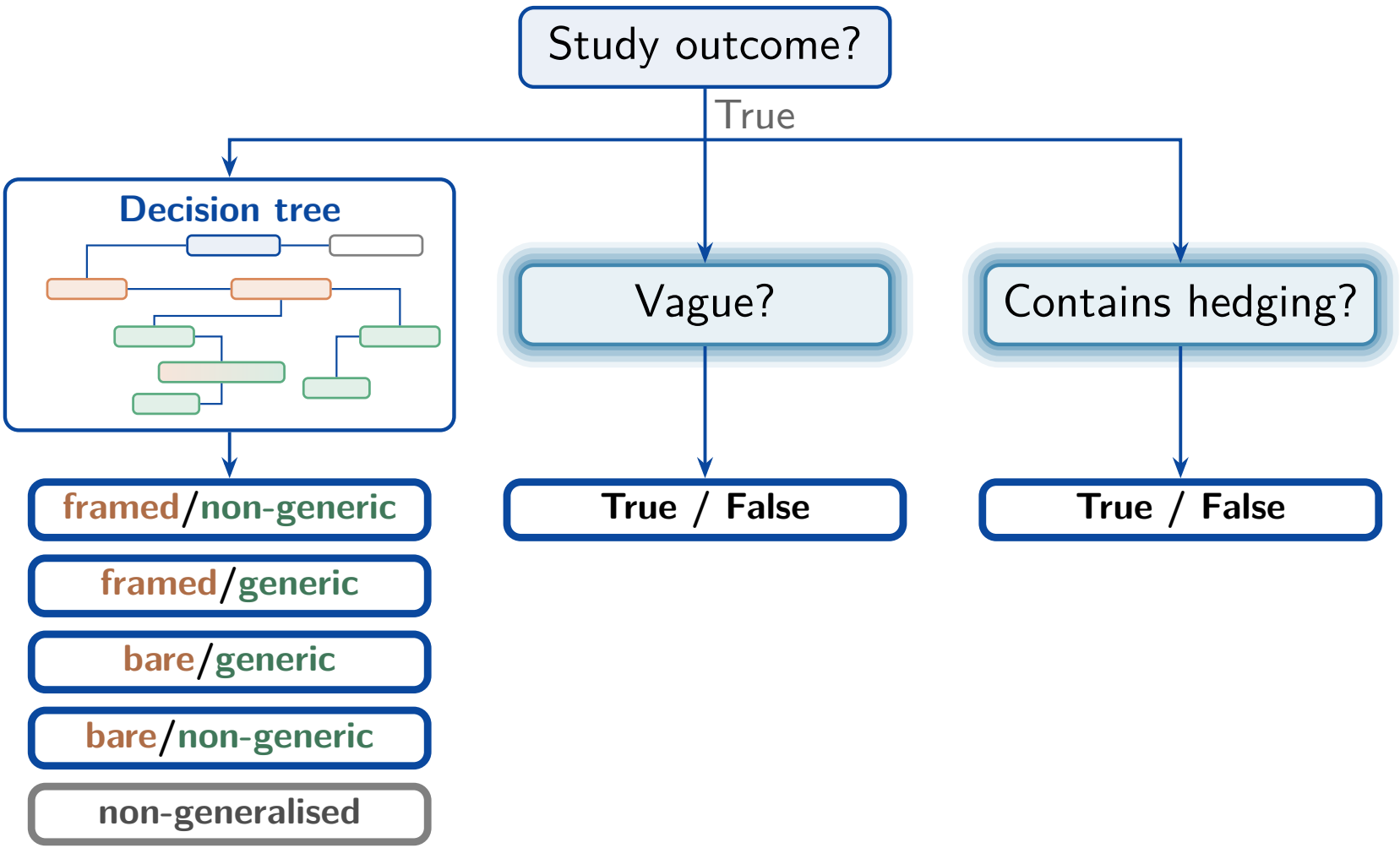
Gating and auxiliary classifiers



Gating and auxiliary classifiers



Gating and auxiliary classifiers



NLPGens: A large labelled generalisation corpus

ACL 2017 | COLING 2020 | ARR 2022 | EMNLP 2023 | EMNLP 2024

∼4,500 NLP papers from NLPeer [Dycke et al., 2023]

↓

∼991K sentences all paper text

↓

∼396K sentences selected sections

↓

180K sentences study outcomes

Human verification shows our labels are reliable

Verification

Is this decision defensible?

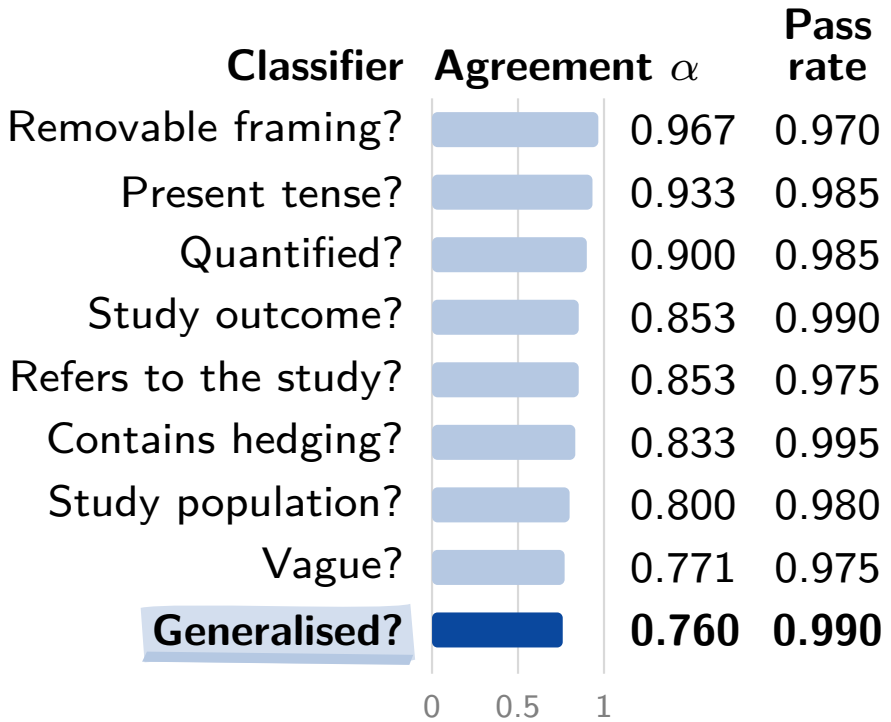
$\alpha = 0.760$

"Generalised?"

inter-annotator agreement

96/100

generalisation labels matched with human judges



Taxonomy **NLPGenX**: five classes of generalisation by framing × genericity

Annotation **NLPGenA**: a decision-tree framework to label at scale

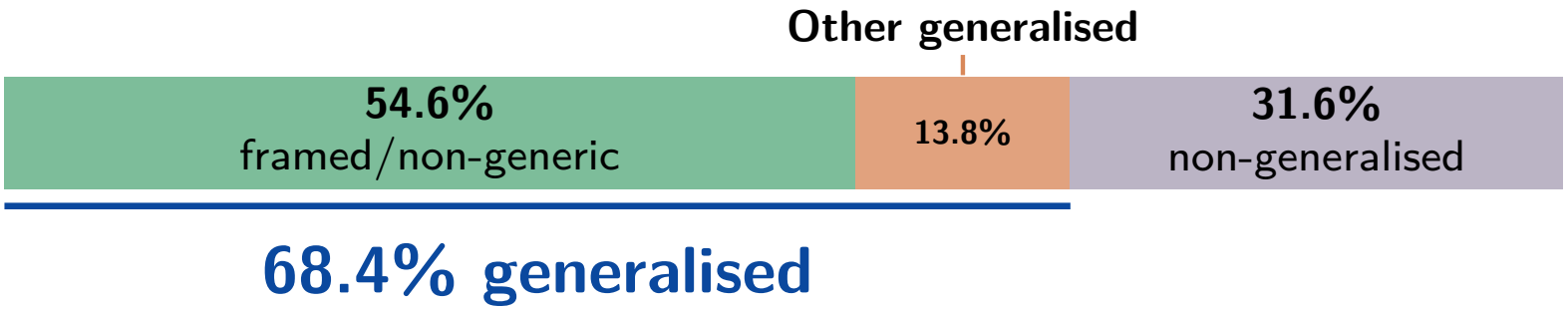
Dataset **NLPGens**: 180K sentences from 4.5K papers

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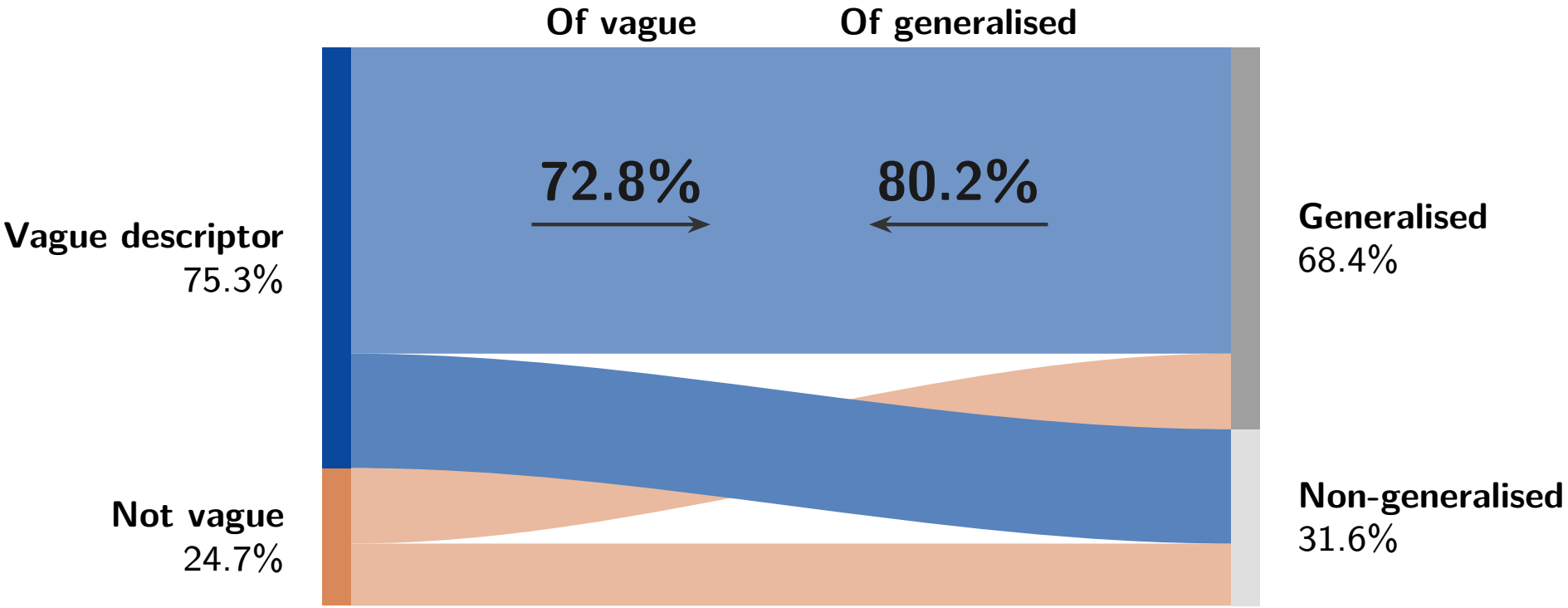
Contributions

Generalisations are common in NLP research

Distribution across all 180,426 outcome sentences

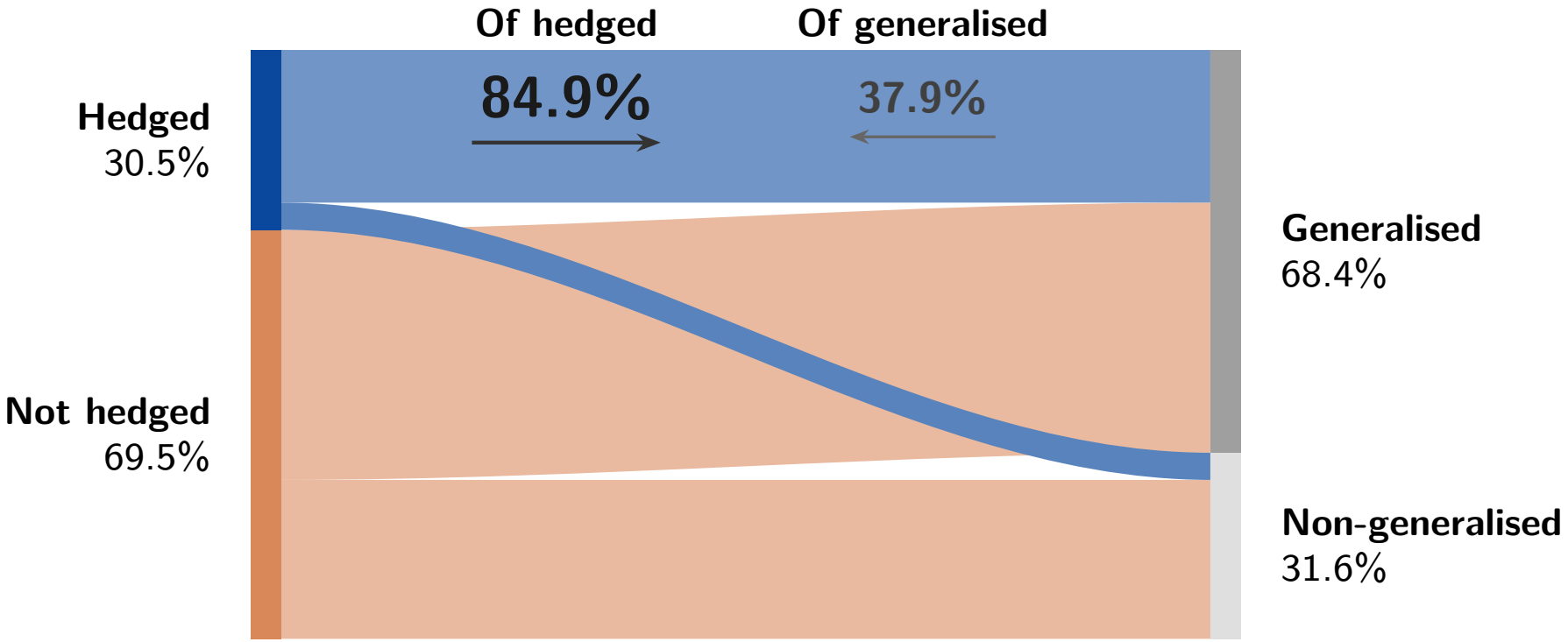


Generalisation is linked with *vagueness*



"LLMs are reliable for annotation."

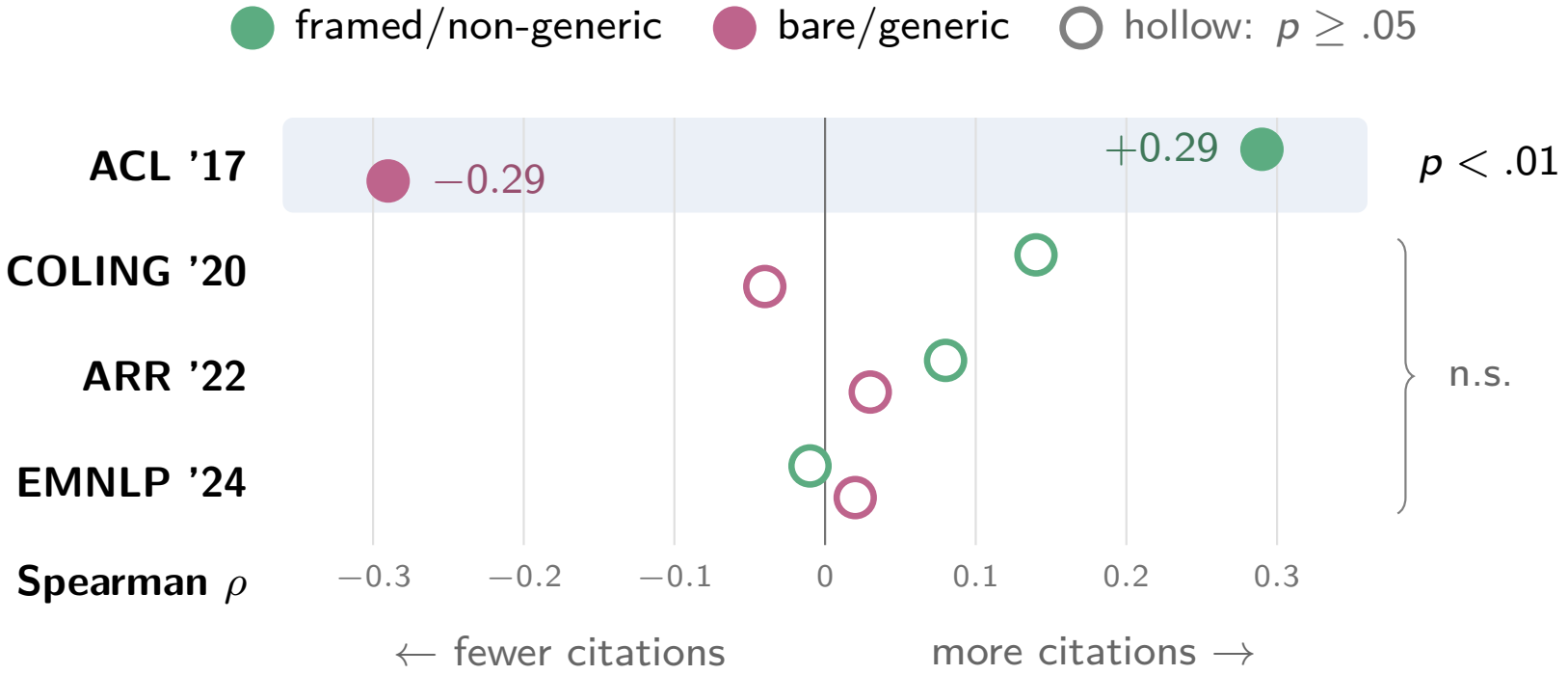
Generalisation often comes with *hedging*



"Results show that `gpt-oss` could produce reliable annotations."

Does generalisation relate to *paper impact*?

Per paper: share of outcome sentences with each label vs. citation count



Only ACL '17 shows a significant link.

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Analysis Generalisation is common, more often when hedged or vague



References I



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